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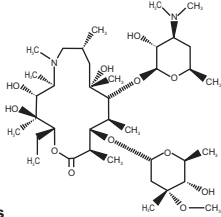
**AZOMAX
(AZITHROMYCIN FOR INJECTION)
Lyophilized**

- 1) Product Name**
AZOMAX Powder for Injection 500 mg/Vial (Azithromycin for Injection)
- 2) Name and Strength of Active Ingredient (s)**
Azithromycin (as Monohydrate) 500 mg/vial

3)Product Description
AZOMAX is a white to off-white lyophilized powder or cake. The reconstituted solution must be clear & colourless solution, should be free from particulate matter. The reconstituted solution is not significantly less clear than an equal volume of purified water contained in a similar vessel and examined similarly.

4)Description
Azithromycin is the first of a class of antibiotics designated chemically as azalides, a subclass of macrolides. Chemically it is derived by insertion of a nitrogen atom into the lactone ring of erythromycin A. The chemical name of Azithromycin is 9-deoxy-9a-aza-9a-methyl-9a-homoerythromycin A. Azithromycin has a chemical formula of C₃₈H₇₂N₂O₉ and a molecular weight of 749.0.

The structural formula is:



5) Pharmacodynamics / Pharmacokinetics

Pharmacodynamic properties

Pharmaco therapeutic group
Azithromycin is Antibacterial agent.

Mechanism of action and Pharmacodynamic effects:

The mode of action of Azithromycin is inhibition of protein synthesis in bacteria by binding to the 50S ribosomal subunit and preventing translocation of peptides.

Pharmacodynamics

Azithromycin is the first of a subclass of macrolides antibiotics, known as azalides, and is chemically different from erythromycin. Chemically it is derived by insertion of a nitrogen atom into the lactone ring of erythromycin A. The mode of action of azithromycin is inhibition of protein synthesis in bacteria by binding to the 50S ribosomal subunit and preventing translocation of peptides.

Azithromycin demonstrates activity in vitro against a wide range of bacteria including:

Gram-positive Aerobic Bacteria

Staphylococcus aureus, *Streptococcus pyogenes* (group A beta-hemolytic streptococci), *Streptococcus pneumoniae*, *alpha-hemolytic streptococci* (viridans group) and other streptococci, and *Corynebacterium diphtheriae*. Azithromycin demonstrates cross-resistance with erythromycin-resistant Gram-positive strains, including *Streptococcus faecalis* (enterococcus) and most strains of methicillin-resistant staphylococci.

Gram-negative Aerobic Bacteria

Haemophilus influenzae, *Haemophilus parainfluenzae*, *Moraxella catarrhalis*, *Acinetobacter* species, *Yersinia* species, *Legionella pneumophila*, *Bordetella pertussis*, *Bordetella parapertussis*, *Shigella* species, *Pasteurella* species, *Vibrio cholerae* and *parahaemolyticus*, *Plesiomonas shigelloides*. Activities against *Escherichia coli*, *Salmonella enteritidis*, *Salmonella typhi*, *Enterobacter* species, *Aeromonas hydrophila* and *Klebsiella* species are variable and susceptibility tests should be performed. *Proteus* species, *Serratia* species, *Morganella* species, and *Pseudomonas aeruginosa* are usually resistant.

Anaerobic Bacteria

Bacteroides fragilis and *Bacteroides* species, *Clostridium perfringens*, *Peptococcus* species and *Peptostreptococcus* species, *Fusobacterium necrophorum* and *Propionibacterium acnes*.

Organisms of Sexually Transmitted Diseases

Azithromycin is active against *Chlamydia trachomatis* and also shows good activity against *Treponema pallidum*, *Neisseria gonorrhoeae*, and *Haemophilus ducreyi*.

Other Organisms

Borrelia burgdorferi (Lyme disease agent), *Chlamydia pneumoniae*, *Mycoplasma pneumoniae*, *Mycoplasma hominis*, *Ureaplasma urealyticum*, *Campylobacter* species and *Listeria monocytogenes*.

Opportunistic Pathogens Associated with HIV Infections

Mycobacterium avium-intracellulare complex, *Pneumocystis carinii* and *Toxoplasma gondii*.

Commonly susceptible species:

Aerobic Gram-positive bacteria:

Staphylococcus aureus, *Streptococcus agalactiae*, Streptococci (Groups C, F, G) and Viridans group streptococci.

Aerobic Gram-negative bacteria:

Bordetella pertussis, *Haemophilus ducreyi*, *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Legionella pneumophila*, *Moraxella catarrhalis* and *Neisseria gonorrhoeae*.

Other

Chlamydia pneumoniae, *Chlamydia trachomatis*, *Mycoplasma pneumoniae* and *Ureaplasma urealyticum*.

Species for which acquired resistance:

Aerobic Gram-positive bacteria: *Streptococcus pneumoniae*, *Streptococcus pyogenes*

Note: Azithromycin demonstrates cross-resistance with erythromycin-resistant gram-positive strains.

Inherently resistant organisms: Enterobacteriaceae, *Pseudomonas*

Pharmacokinetics

Absorption

Following oral administration in humans, azithromycin is widely distributed throughout the body; bioavailability is approximately 37%. The time taken to peak plasma levels is 2-3 hours.

Distribution

In animal studies, high azithromycin concentrations have been observed in phagocytes. In experimental models, higher concentrations of azithromycin are released during active phagocytosis than from non-stimulated phagocytes. In animal models this results in high concentrations of azithromycin being delivered to the site of infection.

Pharmacokinetic studies in humans have shown markedly higher azithromycin levels in tissue than in plasma (up to 50 times the maximum observed concentration in plasma) indicating that the drug is heavily tissue bound. Concentrations in target tissues, such as lung, tonsil and prostate exceed the MIC₉₀ for likely pathogens after a single dose of 500 mg.

Elimination

Plasma terminal elimination half-life closely reflects the tissue depletion half-life of 2 to 4 days. Approximately 12% of an intravenously administered dose is excreted in the urine over 3 days as the parent drug, the majority in the first 24 hours. Biliary excretion of azithromycin is a major route of elimination for unchanged drug following oral administration. Very high concentrations of unchanged drug have been found in human bile, together with 10 metabolites, formed by N- and O-demethylation, by hydroxylation of the desosamine and aglycone rings, and by cleavage of the cladinose conjugate. Comparison of HPLC and microbiological assays in tissues suggests that metabolites play no part in the microbiological activity of azithromycin.

Pharmacokinetics in Special Patient Groups

Elderly

In elderly volunteers (>65 years), slightly higher AUC values were seen after a 5-day regimen than in young volunteers (<40 years), but these are not considered clinically significant, and hence no dose adjustment is recommended.

Renal Impairment

The pharmacokinetics of azithromycin in subjects with mild to moderate renal impairment (GFR 10 - 80 mL/min) were not affected following a single one gram dose of immediate release azithromycin. Statistically significant differences in AUC 0-120 (8.8 µg hr/mL vs. 11.7 µg hr/mL), C_{max} (1.0 µg/mL vs. 1.6 µg/mL) and CL_R (2.3 mL/min/kg vs. 0.2 mL/min/kg) were observed between the group with severe renal impairment (GFR < 10 mL/min) and the group with normal renal function.

Hepatic Impairment

In patients with mild to moderate hepatic impairment, there is no evidence of a marked change in serum pharmacokinetics of azithromycin compared to those with normal hepatic function. In these patients urinary clearance of azithromycin appears to increase, perhaps to compensate for reduced hepatic clearance.

6) Indication

Treatment of community-acquired pneumonia caused by susceptible organisms including *Legionella pneumophila*, in patients who require initial IV therapy. Treatment of pelvic inflammatory disease caused by susceptible organisms (*Chlamydia trachomatis*, *Neisseria gonorrhoea*, *Mycoplasma hominis*), in patients who require initial IV therapy.

7) Recommended Dose and Mode of administration

The dose of Azithromycin for Injection for the treatment of adult patients with community acquired pneumonia is: 500 mg as a single daily intravenous dose for at least two days. Intravenous therapy should be followed by oral therapy of 500 mg Azithromycin administered as a single daily dose to complete a 7 to 10 day course of therapy. The timing of the conversion to oral therapy should be done at the discretion of the physician and in accordance with clinical response.

Treatment of pelvic inflammatory disease due to the indicated organisms:

Recommended Dose: 500 mg as a single daily dose by the IV route for 1-2 days. IV therapy should be followed by Azithromycin by the oral route at a single daily dose of 250 mg to complete a 7-day course of therapy. The timing of the conversion to oral therapy should be done at the discretion of the physician and in accordance with clinical response. If anaerobic microorganism are suspected of contributing to the infection, an antimicrobial anaerobic agent may be administered in combination with azithromycin.

Use in elderly

No dose adjustment is necessary in elderly patients requiring Azithromycin therapy.

Use in patients with renal impairment

No dose adjustment is necessary in patients with mild to moderate (GFR 10 - 80 mL/min) or severe (GFR < 10 mL/min) renal impairment.

Use in patients with hepatic impairment

The same dosage as in patients with normal hepatic function may be used in patients with mild to moderate hepatic impairment.

After re-constitution and dilution, the recommended route of administration for intravenous Azithromycin is by IV infusion only. Do not administer as an intravenous bolus or intramuscular injection.
An IV dose of 500 mg should be infused for a minimum duration of 1 hr.

Reconstitution: Prepare the initial IV solution for infusion by adding 4.8 mL of sterilized water for injection to the 500-mg vial and shaking the vial until all of the drug is dissolved. Since azithromycin IV is supplied under vacuum, it is recommended that a standard 5 mL (non-automated) syringe be used to ensure that the exact amount of 4.8 mL of sterilised water for injections is dispensed. Each mL of reconstituted solution contains 100-mg azithromycin.

Dilute the solution further prior to administration as follows:

Dilution: To provide azithromycin over a concentration range of 1-2 mg/mL, transfer 5 mL of the 100 mg/mL azithromycin solution into the appropriate amount of any of the diluents as follows.

Final infusion solution concentration of 1 and 2 mg/mL requires 500 and 250 mL amount of diluent, respectively.

The reconstituted solution can be diluted with: Normal saline (0.9% sodium chloride); ½ normal saline (0.45% sodium chloride); 5% dextrose in water; lactated Ringer's solution; 5% dextrose in ½ normal saline (0.45% sodium chloride) with 20 mEq KCl; 5% dextrose in lactated Ringer's solution; 5% dextrose in 1/3 normal saline solution (0.3% sodium chloride); 5% dextrose in ½ normal saline (0.45% sodium chloride); Normosol-M in 5% dextrose; Normosol-R in 5% dextrose.

Parenteral drug products should be inspected visually for particulate matter prior to administration. If particulate matter is evident in reconstituted fluids, the drug solution should be discarded.

8) Contraindication

Azithromycin is contraindicated in patients with known hypersensitivity to azithromycin, erythromycin; or any macrolide or ketolide antibiotic, or to any excipients.

9) Warnings and Precautions

Antibiotic associated pseudomembranous colitis has been reported with many antibiotics including Azithromycin. A toxin produced by *Clostridium difficile* appears to be the primary cause. The severity of the colitis may range from mild to life-threatening. It is important to consider this diagnosis in patients who develop diarrhoea or colitis in association with antibiotic use (this may occur up to several weeks after cessation of antibiotic therapy). Mild cases may respond to drug discontinuation alone. However, in moderate to severe cases appropriate therapy with a suitable oral antibacterial agent effective against *Clostridium difficile* should be considered. Fluids, electrolytes and protein replacement should be provided when indicated. Hypertoxin producing strains of C. difficile cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy.

No dose adjustment is needed in patient's with mild or moderate renal impairment (GFR 10-80 mL/min). After oral administration of a single dose of Azithromycin 1 g in subjects with severe renal impairment (GFR < 10 mL/min), mean AUC_{0-120h} and mean C_{max} were increased by approximately 30% and 60%, respectively when compared to subjects with normal renal function. Caution should be exercised when Azithromycin is administered to patients with severe renal impairment.

No dose adjustment is recommended for patients with mild to moderate hepatic impairment. Nonetheless, since liver is the principal route of elimination for Azithromycin, the use of Azithromycin should be undertaken with caution in patients with significant hepatic disease (see PHARMACOKINETICS).

In patients receiving ergot derivatives, ergotism has been precipitated by co administration of some macrolide antibiotics. There are no data concerning the possibility of an interaction between ergot and Azithromycin. However, because of the theoretical possibility of ergotism, Azithromycin and ergot derivatives should not be co administered.

As with any antibiotic preparation, observation for signs of super infection with nonsusceptible organisms, including fungi, is recommended.

Hypersensitivity

As with erythromycin and other macrolides, rare serious allergic reactions, including angioedema and anaphylaxis (rarely fatal), dermatologic reactions including Stevens-Johnson Syndrome (SJS), Toxic Epidermal Necrolysis (TEN) (rarely fatal), and Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) have been reported. Some of these reactions with azithromycin have resulted in recurrent symptoms and required a longer period of observation and treatment.

If an allergic reaction occurs, the drug should be discontinued and appropriate therapy should be instituted. Physicians should be aware that reappearance of the allergic symptoms may occur when symptomatic therapy is discontinued.

Prolongation of the QT interval

Prolonged cardiac repolarization and QT interval, imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in treatment with macrolides, including azithromycin. Prescribers should consider the risk of QT prolongation, which can be fatal, when weighing the risks and benefits of azithromycin for at-risk groups including:

- Patients with congenital or documented QT prolongation
- Patients currently receiving treatment with other active substances known to prolong QT interval, such as antiarrhythmics of Classes IA and III, antipsychotic agents, antidepressants, and fluoroquinolones
- Patients with electrolyte disturbance, particularly in cases of hypokalemia and hypomagnesaemia
- Patients with clinically relevant bradycardia, cardiac arrhythmia or cardiac insufficiency
- Elderly patients: elderly patients may be more susceptible to drug-associated effects on the QT interval

Do not administer Azithromycin for Injection as a bolus or as an intramuscular injection. Reconstitute and dilute the powder for infusion as directed and administer as an intravenous infusion over not less than 60 minutes. All patients who received infusate concentrations above 2.0 mg/mL experienced local infusion site reactions and therefore, higher concentrations should be avoided.

Caution in diabetic patients: 5 mL of reconstituted suspension contain 3.87 g of sucrose.

Due to the sucrose content (3.87 g / 5 mL of reconstituted suspension), this medicinal product is not indicated for persons with fructose intolerance (hereditary fructose intolerance), glucose-galactose malabsorption or saccharase-isomaltase deficiency.

Infantile hypertrophic pyloric stenosis (HPS) has been reported following the use of azithromycin in infants (treatment up to 42 days of life). Parents and caregivers should be informed to contact their physician if vomiting and/or irritability with feeding occurs.

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Steven-Johnson Syndrom (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS)] & acute generalised exanthematous pustulosis (AGEP)], AZOMAX should be discontinued immediately and appropriate treatment should be urgently initiated.

10) Interactions with other medications

Azithromycin does not interact significantly with the hepatic cytochrome P450 system. It is not believed to undergo the pharmacokinetic drug interaction as seen with erythromycin and other macrolides. Hepatic cytochrome P450 induction or inactivation via cytochrome-metabolite complex does not occur with Azithromycin.

Drugs that should not be concomitantly administered with Azithromycin:

Antacids

No effect on overall bioavailability was seen although peak serum concentrations were reduced by approximately 25%. In patients receiving both oral azithromycin and antacids, the drugs should not be taken simultaneously.

Cetirizine

Coadministration of a 5-day regimen of azithromycin with cetirizine 20 mg at steady-state resulted in no pharmacokinetic interaction and no significant changes in the QT interval.

Didanosine (Dideoxyinosine)

Coadministration of 1200 mg/day azithromycin with 400 mg/day didanosine in HIV-positive subjects did not appear to affect the steady-state pharmacokinetics of didanosine as compared with placebo.

Digoxin

Some of the macrolide antibiotics have been reported to impair the microbial metabolism of digoxin in the gut in some patients. In patients receiving concomitant azithromycin, a related azalide antibiotic, and digoxin the possibility of raised digoxin levels should be borne in mind.

Zidovudine

Single 1000 mg doses and multiple 1200 mg or 600 mg doses of azithromycin had little effect on the plasma pharmacokinetics or urinary excretion of zidovudine or its glucuronide metabolites. However, administration of azithromycin increased the concentrations of phosphorylated zidovudine, the clinically active metabolite, in peripheral blood mononuclear cells. The clinical significance of this finding is unclear, but it may be of benefit to patients.

Azithromycin does not interact significantly with the hepatic cytochrome P450 system. It is not believed to undergo the pharmacokinetic drug interactions as seen with erythromycin and other macrolides. Hepatic cytochrome P450 induction or inactivation via cytochrome-metabolite complex does not occur with azithromycin.

Ergot

Due to the theoretical possibility of ergotism, the concurrent use of azithromycin with ergot derivatives is not recommended. Pharmacokinetic studies have been conducted between azithromycin and the following drugs known to undergo significant cytochrome P450 mediated metabolism.

Atorvastatin

Coadministration of atorvastatin (10 mg daily) and azithromycin (500 mg daily) did not alter the plasma concentrations of atorvastatin.

Carbamazepine

No significant effect was observed on the plasma levels of carbamazepine or its active metabolite in patients receiving concomitant azithromycin.

Cimetidine

In a pharmacokinetic study investigating the effects of a single dose of cimetidine, given 2 hours before azithromycin, on the pharmacokinetics of azithromycin, no alteration of azithromycin pharmacokinetics was seen.

Coumarin-Type Oral Anticoagulants

There have been reports of potentiated anticoagulation subsequent to coadministration of azithromycin and coumarin-type oral anticoagulants. Although a causal relationship has not been established, consideration should be given to the frequency of monitoring prothrombin time when azithromycin is used in patients receiving coumarin-type oral anticoagulants.

Cyclosporin

Caution should be exercised before considering concurrent administration of cyclosporin and azithromycin. If coadministration of these drugs is necessary, cyclosporin levels should be monitored and the dose adjusted accordingly.

Efavirenz

Coadministration of a 600 mg single dose of azithromycin and 400 mg efavirenz daily for 7 days did not result in any clinically significant pharmacokinetic interactions.

Fluconazole

Coadministration of a single dose of 1200 mg azithromycin did not alter the pharmacokinetics of a single dose of 800 mg fluconazole. Total exposure and half-life of azithromycin were unchanged by the coadministration of fluconazole, however a clinically insignificant decrease in C_{max} (18%) of azithromycin was observed.

Indinavir

Coadministration of a single dose of 1200 mg azithromycin had no statistically significant effect on the pharmacokinetics of indinavir administered as 800 mg three times daily for 5 days.

Methylprednisolone

Azithromycin had no significant effect on the pharmacokinetics of methylprednisolone.

Midazolam

Coadministration of azithromycin 500 mg/day for 3 days did not cause clinically significant changes in the pharmacokinetics and pharmacodynamics of a single 15 mg dose of midazolam.

Nelfinavir

Coadministration of azithromycin (1200 mg) and nelfinavir at steady state (750 mg three times daily) resulted in increased azithromycin concentrations. No clinically significant adverse effects were observed and no dose adjustment is required.

Rifabutin

Coadministration of azithromycin and rifabutin did not affect the serum concentrations of either drug. Neutropenia was observed in subjects receiving concomitant treatment of azithromycin and rifabutin. Although neutropenia has been associated with the use of rifabutin, a causal relationship to combination with azithromycin has not been established.

Sildenafil

There was no evidence of an effect of azithromycin (500mg daily for 3 days) on the AUC and C_{max} of sildenafil or its major circulating metabolite.

Terfenadine

Pharmacokinetic studies have reported no evidence of an interaction between azithromycin and terfenadine. There have been rare cases reported where the possibility of such an interaction could not be entirely excluded; however, there was no specific evidence that such an interaction had occurred.

Theophylline

There is no evidence of a clinically significant pharmacokinetic interaction when azithromycin and theophylline are co-administered to healthy volunteers.

Triazolam

Coadministration of azithromycin with 0.125 mg triazolam had no significant effect on any of the pharmacokinetic variables for triazolam compared to triazolam and placebo.

Trimethoprim/sulfamethoxazole

Coadministration of trimethoprim/sulfamethoxazole DS (160 mg/800 mg) for 7 days with azithromycin 1200 mg on Day 7 had no significant effect on peak concentrations, total exposure or urinary excretion of either trimethoprim or sulfamethoxazole. Azithromycin serum concentrations were similar to those seen in other studies.

11) Pregnancy and Lactation

Use in Pregnancy:

CATEGORY B1

Studies in mice and rats have demonstrated that Azithromycin crosses the placenta. Following an oral dose of 200 mg/kg/day, Azithromycin concentrations in mouse and rat foetal tissue homogenates were 5 to 10 fold higher than corresponding maternal plasma concentrations. No animal studies of embryofoetal development have been conducted by the IV route. Azithromycin was not foetotoxic or teratogenic in mice and rats at oral doses that were moderately maternotoxic. Plasma levels for Azithromycin were lower than the clinical C_{max} in both species at the high dose of 200 mg/kg/day. Azithromycin powder for solution for infusion should only be used in pregnant women where adequate alternatives are not available.

Use in Lactation:

There are no data on the possible secretion of Azithromycin into animal or human breast milk. Azithromycin should only be used in lactating women where adequate alternatives are not available.

12) Undesirable Effects

Clinical side effects leading to discontinuations from these studies were most commonly gastrointestinal (abdominal pain, nausea, vomiting, diarrhoea), and rashes; laboratory side effects leading to discontinuation were increases in transaminase levels and/or alkaline phosphatase levels.

Post-marketing experience:

Cardiac Disorders: Palpitations and arrhythmias including ventricular tachycardia have been reported. There have been rare reports of QT prolongation and torsades de pointes (see Warnings and Precautions).

Skin and Subcutaneous Tissue Disorders: Allergic reactions including pruritus, rash, photosensitivity, edema, urticarial, and angioedema. Rarely, serious cutaneous adverse reactions including erythema multiforme, SJS, TEN and DRESS have been reported.

Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)].

Gastrointestinal Disorders: Infantile hypertrophic pyloric stenosis

13) Overdose and treatment

Most adverse events experienced in higher than recommended doses were similar to those in type and may be more frequent than those seen at normal doses. The incidence of tinnitus and ototoxicity is more frequent in overdose than at normal doses. In the event of overdose, general symptomatic and supportive measures are indicated as required.

As with many cationic: amphiphilic drugs, phospholipidosis has been observed in some tissues of mice, rats and dogs given multiple doses of Azithromycin. It has been demonstrated in numerous organ systems in dogs administered doses which, based on pharmacokinetics, are as low as 2-3 times greater than the recommended human dose and in rats at doses comparable to the human dose. This effect is reversible after cessation of Azithromycin treatment. The significance of these findings for humans with overdose of Azithromycin is unknown.

14) Storage Condition

Azithromycin for Injection should be stored below 30°C. After reconstitution solution can be stored below 30°C for not more than 24 hours. Diluted solution can be stored below 30°C for not more than 24 hours and at 2° C to 8° C for not more than 7 days.

15) Dosage Forms and Packaging available

Stable powder for injection in 10 mL Clear glass Lyo V